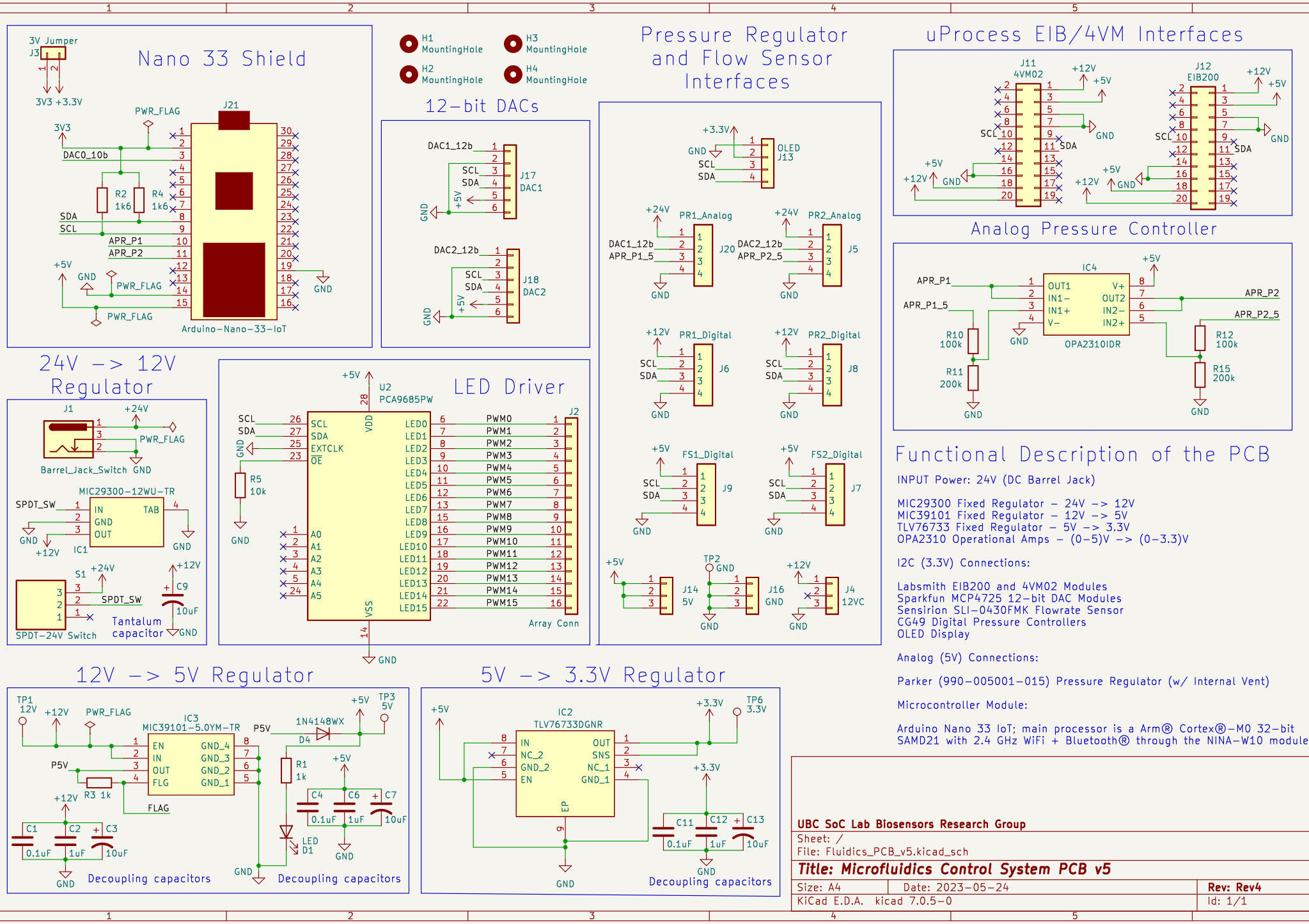

Nano 33 Shield
3V Jumper
J3
3V3 +3.3V
PWR_FLAG
J21
3V3
DAC0_10b
R2 1k6
R4 1k6
SDA
SCL
APR_P1
APR_P2
+5V
GND
PWR_FLAG
Arduino-Nano-33-IoT
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
12-bit DACs
DAC1_12b
J17 DAC1
DAC2_12b
J18 DAC2
Pressure Regulator and Flow Sensor Interfaces
+3.3V
OLED J13
PR1_Analog
J20
DAC1_12b
APR_P1_5
PR2_Analog
J5
DAC2_12b
APR_P2_5
PR1_Digital
J6
PR2_Digital
J8
FS1_Digital
J9
FS2_Digital
J7
J14 5V
TP2
J16 GND
J4 12VC
uProcess EIB/4VM Interfaces
J11 4VM02
J12 EIB200
+12V
+5V
GND
SCL
SDA
Analog Pressure Controller
IC4
OPA2310IDR
APR_P1
APR_P1_5
APR_P2
APR_P2_5
R10 100k
R11 200k
R12 100k
R15 200k
24V -> 12V Regulator
J1
Barrel_Jack_Switch
MIC29300-12WU-TR
SPDT_SW
IC1
S1
SPDT-24V Switch
C9 10uF
Tantalum capacitor
LED Driver
U2 PCA9685PW
R5 10k
Array Conn
Functional Description of the PCB
INPUT Power: 24V (DC Barrel Jack)
MIC29300 Fixed Regulator - 24V -> 12V
MIC39101 Fixed Regulator - 12V -> 5V
TLV76733 Fixed Regulator - 5V -> 3.3V
OPA2310 Operational Amps - (0-5)V -> (0-3.3)V
I2C (3.3V) Connections:
Labsmith EIB200 and 4VM02 Modules
Sparkfun MCP4725 12-bit DAC Modules
Sensirion SLI-0430FMK Flowrate Sensor
CG49 Digital Pressure Controllers
OLED Display
Analog (5V) Connections:
Parker (990-005001-015) Pressure Regulator (w/ Internal Vent)
Microcontroller Module:
Arduino Nano 33 IoT; main processor is a Arm® Cortex®-M0 32-bit SAMD21 with 2.4 GHz WiFi + Bluetooth® through the NINA-W10 module
12V -> 5V Regulator
TP1 12V
IC3 MIC39101-5.0YM-TR
P5V
1N4148WX
D4
R3 1k
R1 1k
FLAG
LED D1
Decoupling capacitors
5V -> 3.3V Regulator
IC2 TLV76733DGNR
TP6 3.3V
Decoupling capacitors
UBC SoC Lab Biosensors Research Group
Sheet: /
File: Fluidics_PCB_v5.kicad_sch
Title: Microfluidics Control System PCB v5
Size: A4
Date: 2023-05-24
Rev: Rev4
KiCad E.D.A. kicad 7.0.5-0
Id: 1/1